Experiment 12

Experiment to detect outliers in a given data set

Aim:

To detect outliers in a given data set.

Code:

```
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt

array = np.array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
array

array.mean()

np.percentile(array, 25)
np.percentile(array, 50)
np.percentile(array, 75)
np.percentile(array, 100)

def outDetection(array):
    sorted(array)
    Q1, Q3 = np.percentile(array, [25, 75])
    IQR = Q3 - Q1
    lr = Q1 - (1.5 * IQR)
    ur = Q3 + (1.5 * IQR)
    return lr, ur

lr, ur = outDetection(array)
lr, ur

sns.displot(array)
sns.distplot(array)

new_array = array[(array > lr) & (array < ur)]
new_array

sns.displot(new_array)

lr1, ur1 = outDetection(new_array)
lr1, ur1

final_array = new_array[(new_array > lr1) & (new_array < ur1)]
final_array

sns.distplot(final_array)
```

Output:

```
In [1]: array = np.array([27, 50, 44, 6, 58, 61, 23, 86, 67, 20, 75, 7, 79, 61, 90, 54])
In [2]: array.mean()
Out[2]: 50.5

In [3]: np.percentile(array, 25)
Out[3]: 26.0

In [4]: np.percentile(array, 50)
Out[4]: 56.0

In [5]: np.percentile(array, 75)
Out[5]: 69.0

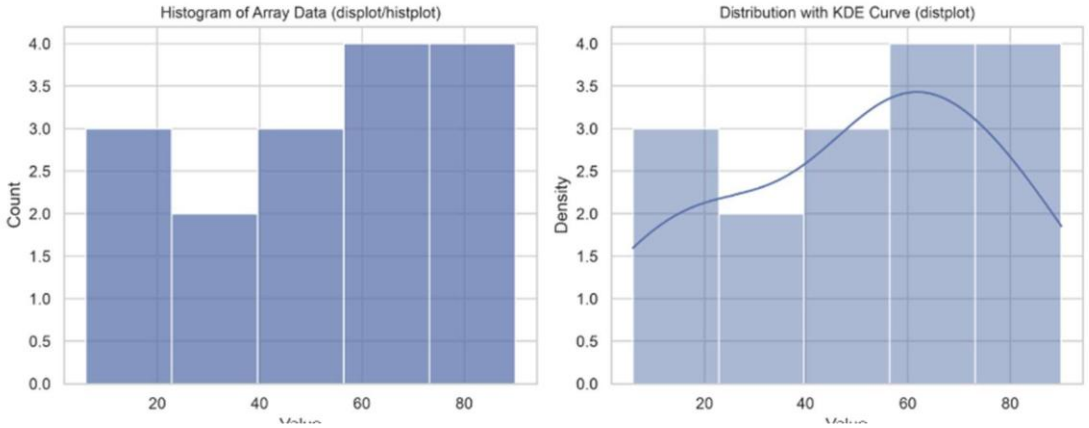
In [6]: np.percentile(array, 100)
Out[6]: 90.0

In [7]: lr, ur = outDetection(array)
In [8]: lr, ur
Out[8]: (-38.5, 133.5)

In [9]: new_array = array[(array > lr) & (array < ur)]
In [10]: new_array
Out[10]: array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])

In [11]: lr1, ur1 = outDetection(new_array)
In [12]: lr1, ur1
Out[12]: (-38.5, 133.5)

In [13]: final_array = new_array[(new_array > lr1) & (new_array < ur1)]
In [14]: final_array
Out[14]: array([27, 50, 44,  6, 58, 61, 23, 86, 67, 20, 75,  7, 79, 61, 90, 54])
```



Result:

Outlier detection using the IQR method was successfully performed. Outliers beyond the bounds [Q1 - 1.5*IQR, Q3 + 1.5*IQR] were identified and removed from the dataset. Box plots before and after filtering visually confirmed the elimination of extreme values